AssignmentNumber:7.3

Task 1: Fixing Syntax Errors

Scenario

You are reviewing a Python program where a basic function definition contains a syntax error.

```python
def add(a, b)
    return a + b
```

Requirements

• Provide a Python function add(a, b) with a missing colon

• Use an AI tool to detect the syntax error

• Allow AI to correct the function definition

• Observe how AI explains the syntax issue

Expected Output

• Corrected function with proper syntax

• Syntax error resolved successfully

• AI-generated explanation of the fix.

CODE:

```
AIAC_LAB 7.3.py
def add(a ,b)
    return a + b
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.2.py"
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.2.py", line 1
    def add(a ,b)
                ^
SyntaxError: expected ':'
PS C:\AI Assissted Coding Assessments>
```

Corrected code:

```
AIAC_LAB 5.3.py   AIAC_LAB 6.2.py   AIAC_LAB 7.3.py
AIAC_LAB 7.3.py > add
def add(a ,b):
    return a + b
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.2.py"
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.2.py", line 1
    def add(a ,b)
                ^
SyntaxError: expected ':'
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
PS C:\AI Assissted Coding Assessments>
```

Explanation:

Here after declaring the function we forgotted the colon . so it has raised the syntax error.

Task 2: Debugging Logic Errors in Loops

Scenario

You are debugging a loop that runs infinitely due to a logical mistake.

```python
def count_down(n):
    while n >= 0:
        print(n)
        n += 1  # Should be n -= 1
```

Requirements

- Provide a loop with an increment or decrement error
- Use AI to identify the cause of infinite iteration
- Let AI fix the loop logic
- Analyze the corrected loop behavior

Expected Output

- Infinite loop issue resolved
- Correct increment/decrement logic applied
- AI explanation of the logic error

Code:

```python
    return a + b
'''
def count_down(n):
    while n >= 0:
        print(n)
        n+=1
number = int(input("Enter a number to count down from: "))
count_down(number)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

```
144937
144938
144939
144940
144941
144942
144943
144944
144945
144946
144947
144948
144949
144950
144951
144952
144953
```

Corrected code:

```
    return a + b
'''
def count_down(n):
    while n >= 0:
        print(n)
        n-=1
number = int(input("Enter a number to count down from: "))
count_down(number)
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER        Python

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter a number to count down from: 5
5
4
3
2
1
0
PS C:\AI Assissted Coding Assessments>
```

Explanation:

While n is incremented by one it is going to the infinite loop condition is not failed .it continues iteration.then I have chanfe to -1 the condition fails and not going to the infinite loop.

Task 3: Handling Runtime Errors (Division by Zero)

Scenario

A Python function crashes during execution due to a division by zero error.

```
# Debug the following code
def divide(a, b):
    return a / b

print(divide(10, 0))
```

Requirements

- Provide a function that performs division without validation
- Use AI to identify the runtime error
- Let AI add try-except blocks for safe execution
- Review AI's error-handling approach

Expected Output

- Function executes safely without crashing
- Division by zero handled using try-except
- Clear AI-generated explanation of runtime error handling.

Code:

```
def divide(a, b):
    return a / b
divide(10,0)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter a number to count down from: 5
5
4
3
2
1
0
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 12, in <module>
    divide(10,0)
    ~~~~~~^^^^^^
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 11, in divide
    return a / b
           ~~^~~
ZeroDivisionError: division by zero
```

Corrected code:

```
count_down(number)'''
def divide(a ,b):
    return a / b
try:
    num1 = float(input("Enter the first number: "))
    num2 = float(input("Enter the second number: "))
    result = divide(num1, num2)
    print(f"The result of dividing {num1} by {num2} is: {result}")
except ZeroDivisionError:
    print("Error: Cannot divide by zero.")
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER
3
2
1
0
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 12, in <module>
    divide(10,0)
    ~~~~~~^^^^^^
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 11, in divide
    return a / b
           ~~^~~
ZeroDivisionError: division by zero
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter the first number: 10
Enter the second number: 0
Error: Cannot divide by zero.
PS C:\AI Assissted Coding Assessments>
```

Explanation:

First it has raised the zero division error and after using the try and except handle it has turned to error raised.

Task 4: Debugging Class Definition Errors

Scenario

You are given a faulty Python class where the constructor is incorrectly defined.

```python
class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

Requirements

• Provide a class definition with missing self-parameter

• Use AI to identify the issue in the __init__() method

• Allow AI to correct the class definition

• Understand why self is required

Expected Output

• Corrected __init__() method

• Proper use of self in class definition

• AI explanation of object-oriented error.

Code:

```python
class rectancle:
    def __init__(length, breadth):
        self.length = length
        self.breadth = breadth
# Example usage
length = float(input("Enter the length of the rectangle: "))
breadth = float(input("Enter the breadth of the rectangle: "))
rectangle = rectancle(length, breadth)
print(f"Area of the rectangle: {rectangle.length * rectangle.breadth}")
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2 Python

```
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter the length of the rectangle: 5
Enter the breadth of the rectangle: 6
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 26, in <module>
    rectangle = rectancle(length, breadth)
TypeError: rectancle.__init__() takes 2 positional arguments but 3 were given
PS C:\AI Assissted Coding Assessments>
```

Corrected Code:

```python
    print("Error: Cannot divide by zero.")'''
class rectancle:
    def __init__(self,length, breadth):
        self.length = length
        self.breadth = breadth
# Example usage
length = float(input("Enter the length of the rectangle: "))
breadth = float(input("Enter the breadth of the rectangle: "))
rectangle = rectancle(length, breadth)
print(f"Area of the rectangle: {rectangle.length * rectangle.breadth}")
```

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER 2        Python
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter the length of the rectangle: 5
Enter the breadth of the rectangle: 6
Area of the rectangle: 30.0
PS C:\AI Assissted Coding Assessments>
```

Explanation:

I have missed self parameter in the function . so It has raised the error .After replaceing the self parameter there is no error.

Task 5: Resolving Index Errors in Lists

Scenario

A program crashes when accessing an invalid index in a list.

> A program crashes when accessing an invalid index in a list.
>
> python
>
> ```python
> numbers = [1, 2, 3]
> print(numbers[5])
> ```

Requirements

- Provide code that accesses an out-of-range list index
- Use AI to identify the Index Error
- Let AI suggest safe access methods
- Apply bounds checking or exception handling

Expected Output

- Index error resolved
- Safe list access logic implemented
- AI suggestion using length checks or exception handling.

Code:

```
number = [1,2,3]
print(number[6])
```

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER 2
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter the length of the rectangle: 5
Enter the breadth of the rectangle: 6
Area of the rectangle: 30.0
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 29, in <module>
    print(number[6])
          ~~~~~~^^^
IndexError: list index out of range
PS C:\AI Assissted Coding Assessments>
```

Corrected code:

```
print(f"Area of the rectangle: {rectangle.length * rectangle.breadth}")'''
number = [1,2,3]
print(number[0])
```

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER 2
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Enter the length of the rectangle: 5
Enter the breadth of the rectangle: 6
Area of the rectangle: 30.0
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 29, in <module>
    print(number[6])
          ~~~~~~^^^
IndexError: list index out of range
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
Traceback (most recent call last):
  File "c:\AI Assissted Coding Assessments\AIAC_LAB 7.3.py", line 29, in <module>
    print(number[3])
          ~~~~~~^^^
IndexError: list index out of range
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 7.3.py"
1
PS C:\AI Assissted Coding Assessments>
```

Explanation:

I have declared three numbers.but I am calling the number beyond the numbers so the error raised like list out of index.

After I have change to 0 the output as 1.